

Rule-2: Right Hand side $x(x)$ of the differential equation $f(D)y = x(x)$ is an expression of the form e^{ax} .

i. $\frac{1}{f(D)} e^{ax} = \frac{e^{ax}}{f(a)}$ When $f(a) \neq 0$.

ii. $\frac{1}{f(D)} e^{ax} = x \cdot \frac{1}{f'(D)} e^{ax}$ When $f(a) = 0$

$= x \cdot \frac{1}{f'(a)} e^{ax}$ When $f'(a) \neq 0$

iii. $\frac{1}{f(D)} e^{ax} = x^2 \cdot \frac{1}{f''(D)} e^{ax}$ When $f(a) = 0$

$= x^2 \cdot \frac{1}{f''(a)} e^{ax}$ When $f'(a) = 0$

$= x^2 \cdot \frac{1}{f''(a)} e^{ax}$ When $f''(a) \neq 0$

Q. solve the differential equations.

i. $(D^2 - 3D + 2)y = e^{5x}$

ii. $(D^2 - a^2)y = e^{ax}$

iii. $(D^3 + 3D^2 + 3D + 1)y = e^{-x}$

iv. $(D^2 + 6D + 5)y = 2e^x + 10e^{5x}$

v. $(D^2 + 4D + 4)y = 3e^{-x}$ when $y(0) = 3$ and $y'(0) = 1$

i. solution: Given that, $(D^2 - 3D + 2)y = e^{5x}$

$$\therefore \text{A.E.: } m^2 - 3m + 2 = 0$$

$$\Rightarrow m = 1, 2$$

$$\therefore y_c = C_1 e^x + C_2 e^{2x}$$

$$\text{Again, } y_p = \frac{1}{D^2 - 3D + 2} e^{5x}$$

$$= \frac{1}{5^2 - 3 \cdot 5 + 2} e^{5x} = 0$$

$$= \frac{1}{12} e^{5x}$$

$\therefore$ The general solution is, $y = y_c + y_p$

$$= C_1 e^x + C_2 e^{2x} + \frac{1}{12} e^{5x}$$

Where C_1 & C_2 are two arbitrary constant.

ii. solution: The given equation is, $(D^2 - a^2)y = e^{ax}$

$$\therefore \text{A.E.: } m^2 - a^2 = 0$$

$$\Rightarrow m = \pm a$$

$$\therefore y_c = C_1 e^{-ax} + C_2 e^{ax}$$

$$\text{Again, } y_p = \frac{1}{D^2 - a^2} e^{ax}$$

$$= x \cdot \frac{1}{2D} e^{ax}$$

$$= \frac{1}{2a} x e^{ax}$$

The general solution is, $y = y_c + y_p$

$$\therefore y = C_1 e^{-ax} + C_2 e^{ax} + \frac{1}{2a} x e^{ax}$$

iii. Solution: The given equation is,

$$(D^3 + 3D^2 + 3D + 1)y = e^{-x}$$

$$\therefore \text{A.E.} : m^3 + 3m^2 + 3m + 1 = 0$$

$$\Rightarrow m^2(m+1) + 2m^2 + 3m + 1 = 0$$

$$\Rightarrow m^2(m+1) + 2m(m+1) + (m+1) = 0$$

$$\Rightarrow (m+1)(m^2 + 2m + 1) = 0$$

$$\therefore m = -1, -1, -1$$

$$\text{C.F.} : y_c = (C_1 + C_2 x + C_3 x^2) e^{-x}$$

$$\text{P.I.} : y_p = \frac{1}{D^3 + 3D^2 + 3D + 1} \cdot e^{-x}$$

$$= \frac{1}{(-1)^3 + 3(-1)^2 + 3(-1) + 1} \cdot e^{-x}$$

$$= \frac{1}{0} e^{-x}$$

$$\therefore y_p = x \cdot \frac{1}{3D^2 + 6D + 3} e^{-x}$$

$$= x^2 \cdot \frac{1}{6D + 6} e^{-x}$$

$$= x^3 \cdot \frac{1}{6} \cdot e^{-x}$$

$\therefore$ The general solution is, $y = y_c + y_p$

$$= (C_1 + C_2 x + C_3 x^2) e^{-x} + \frac{1}{6} x^3 e^{-x}$$

iv. Solution: The given equation is,

$$(D^2 + 6D + 5)y = 2e^x + 10e^{5x}$$

$$\therefore \text{A.E.: } m^2 + 6m + 5 = 0$$

$$\Rightarrow m = -1, -5$$

$$\text{C.F.: } y_c = C_1 e^{-x} + C_2 e^{-5x}$$

$$\text{P.I.: } y_p = \frac{1}{D^2 + 6D + 5} \cdot (2e^x + 10e^{5x})$$

$$= 2 \cdot \frac{1}{1^2 + 6 \cdot 1 + 5} e^x + 10 \cdot \frac{1}{5^2 + 6 \cdot 5 + 5} e^{5x}$$

$$= \frac{1}{6} (e^x + e^{5x})$$

$$\therefore \text{G.S.: } y = y_c + y_p = C_1 e^{-x} + C_2 e^{-5x} + \frac{1}{6} (e^x + e^{5x})$$

v. Solution: The given equation is,

$$(D^2 + 4D + 4)y = 3e^{-x} \text{ when } y(0) = 3 \text{ and } y'(0) = 1$$

$$\therefore \text{A.E.: } y_c = (C_1 + C_2 x) e^{-2x}$$

$$\text{P.I.: } y_p = \frac{1}{D^2 + 4D + 4} \cdot 3e^{-x}$$

$$= \frac{1}{(-1)^2 + 4(-1) + 4} \cdot 3e^{-x}$$

$$= 3e^{-x}$$

$$\text{G.S.: } y = y_c + y_p = (C_1 + C_2 x) e^{-2x} + 3e^{-x}$$

$$\text{When } y(0) = 3, \text{ then } C_1 = 0$$

$$\text{Again, } y'(x) = C_2 e^{-2x} - 2(C_1 + C_2 x) e^{-2x} - 3e^{-x}$$

$$\text{When, } y'(0) = 1, \text{ then } C_2 - 2C_1 - 3 = 1 \Rightarrow C_2 = 4$$

$$\therefore y = 4x e^{-2x} + 3e^{-x} \quad \underline{\underline{Ans}}$$

Rule-3: Right Hand side $x(x)$ of the differential equation $f(D)y = x(x)$ in terms of $\sin ax$ or $\cos ax$.

$$* f(D^r) \sin(ax+b) = f(-a^r) \sin(ax+b)$$

$$* f(D^r) \cos(ax+b) = f(-a^r) \cos(ax+b)$$

a. solve the differential equations

$$i. (D^2+1)y = \cos 2x$$

$$ii. (D^2-3D+2)y = \sin 3x$$

$$iii. (D^2+1)y = \sin x$$

$$iv. (D^2+1)y = \cos x$$

$$v. (D^2+1)y = \sin x \sinh 2x$$

$$vi. (D^2+4)y = \cos 2x$$

$$vii. (D^2+4)y = \sin 2x$$

i. solution: Given that, $(D^2+1)y = \cos 2x$

$$\therefore \text{A.E.} : m^2+1=0$$

$$\Rightarrow m = \pm i$$

$$\text{C.F.} : y_c = A \cos x + B \sin x$$

$$\text{Now, P.I.} : y_p = \frac{1}{D^2+1} \cos 2x$$

$$= \frac{1}{-2^2+1} \cos 2x = -\frac{1}{3} \cos 2x$$

$$\therefore y = y_c + y_p = A \cos x + B \sin x - \frac{1}{3} \cos 2x$$

iii. Solution: Given that, $(D^2+1)y = \sin x$

A.E.: $m^2+1=0$

$\Rightarrow m = \pm i$

C.F.: $y_c = A \cos x + B \sin x$

P.I.: $y_p = \frac{1}{D^2+1} \sin x$

$= x \cdot \frac{1}{2D} \sin x$

$= -\frac{x}{2} \cos x$

$\therefore$ General solution, $y = y_c + y_p$
 $= A \cos x + B \sin x - \frac{x}{2} \cos x$

v. solution: Given that, $(D^2+1)y = \sin x \sin 2x$

A.E.: $m^2+1=0$

$\Rightarrow m = \pm i$

C.F.: $y_c = A \cos x + B \sin x$

P.I.: $y_p = \frac{1}{D^2+1} \sin x \sin 2x$

$= \frac{1}{2} \cdot \frac{1}{D^2+1} (\cos x - \cos 3x)$

$= \frac{1}{2} \cdot x \cdot \frac{1}{2D} \cos x - \frac{1}{2} \cdot \frac{1}{-3^2+1} \cos 3x$

$= \frac{x}{4} \sin x + \frac{1}{16} \cos 3x$

$\therefore$ General solution, $y = y_c + y_p$

$= A \cos x + B \sin x + \frac{x}{4} \sin x + \frac{1}{16} \cos 3x$

Rule-4: $f(D) e^{ax} \cdot x^n = e^{ax} f(D+a) x^n$

$0 = (s-m) m = 0 = m$ $\downarrow$ using rule-1

Rule-5: $f(D) e^{ax} \sin bx \text{ or } \cos bx = e^{ax} f(D+a) \sin bx \text{ or } \cos bx$

$\downarrow$ using rule-3

Rule-6: $f(D) x^n \sin bx \text{ or } \cos bx = f(D) x^n \sin bx$
 $= f(D) x^n \left(\frac{e^{ib} - e^{-ib}}{2i} \right)$

using rule-4 $\rightarrow$ using rule-1

Rule-7: $f(D) x^n \cos bx = f(D) x^n \left(\frac{e^{ib} + e^{-ib}}{2} \right)$

using rule-4

$\downarrow$ using rule-1

Q. solve the differential equations

i. $(D^2 - 2D) y = e^x \sin x$

ii. $(D^2 - 9) y = e^{3x} \cos x$

iii. $(D^2 + D) y = x \cos x$

iv. $(D^2 - 1) y = x^2 \cos x$

v. $(D^2 - 1) y = x e^{2x}$

vi. $(D^2 - 4D + 4) y = 3x^2 e^{2x} \sin 2x$

i. solution: Given that, $(D^2 - 2D)y = e^{2x} \sin x$

$$\text{A.E.: } m^2 - 2m = 0 \Rightarrow m(m-2) = 0$$

$$\Rightarrow m = 0, 2$$

$$\therefore \text{C.F.: } y_c = A + Be^{2x}$$

$$\text{P.I.: } y_p = \frac{1}{D^2 - 2D} e^{2x} \sin x$$

$$= e^{2x} \frac{1}{(D+1)^2 - 2(D+1)} \sin x$$

$$= e^{2x} \frac{1}{D^2 - 1} \sin x$$

$$= e^{2x} \cdot \frac{1}{-1^2 - 1} \sin x$$

$$= -\frac{1}{2} e^{2x} \sin x$$

$$\text{G.S.: } y = y_c + y_p = A + Be^{2x} - \frac{1}{2} e^{2x} \sin x$$

ii. solution: Given that, $(D^2 - 9)y = e^{3x} \cos x$

$$\text{A.E.: } m^2 - 9 = 0 \Rightarrow m = \pm 3$$

$$\therefore \text{C.F.: } y_c = Ae^{3x} + Be^{-3x}$$

$$\text{P.I.: } y_p = \frac{1}{D^2 - 9} e^{3x} \cos x$$

$$= \frac{e^{3x} \cdot 1}{(D+3)^2 - 9} \cos x$$

$$= e^{3x} \cdot \frac{1}{D^2 + 6D} \cos x$$

$$= e^{3x} \cdot \frac{1}{-1^2 + 6D} \cos x$$

$$= e^{3x} \cdot \frac{6D+1}{36D^2-1} \cos x$$

$$= e^{3x} \cdot \frac{6D+1}{36(-1)^2-1} \cos x$$

$$= -\frac{1}{37} e^{3x} (6 \sin x + \cos x)$$

$$= -\frac{1}{37} e^{3x} (\cos x - 6 \sin x)$$

$$\therefore \text{G.S.: } y = y_c + y_p = A e^{+3x} + B e^{3x} - \frac{1}{37} e^{3x} (\cos x - 6 \sin x)$$

iii. solution: Given that $(D^2+D)y = x \cos x$

$$\therefore \text{A.E.: } m^2+m=0$$

$$\Rightarrow m(m+1)=0$$

$$\therefore m=0, -1$$

$$\text{C.F.: } y_c = A + B e^{-x}$$

$$\text{P.I.: } y_p = \frac{1}{D^2+D} x \cos x$$

$$\text{Let, } x = \frac{1}{D^2+D} x \cos x$$

$$\text{and } y = \frac{1}{D^2+D} x \sin x$$

$$\text{Now, } x + iy = \frac{1}{D^2+D} \cdot x \cdot e^{ix}$$

$$= e^{ix} \frac{1}{(D+i)^2 + (D+i)}$$

$$= e^{ix} \cdot \frac{1}{D^2+2iD+D+i}$$

$$= e^{ix} \cdot \frac{1}{D^2 + 3iD + D - 1 + i} x$$

$$= e^{ix} \frac{1}{-1 \{ 1 - (D^2 + 3iD + D) + i \}} x$$

$$= -e^{ix} \{ 1 - (D^2 + 3iD + D) + i \}^{-1} x$$

$$= -e^{ix} \left\{ x + \frac{D^2 + 3iD + D + i}{-1} x \right\}$$

$$= -e^{ix} (ix + 3i + 1)$$

$$= -(\cos x + i \sin x) (ix + 3i + 1)$$

$$= -\left[\cos x - (x+3) \sin x \right] + i \left[\sin x + (x+3) \cos x \right]$$

Equating real part on both sides,

$$X = \frac{1}{D^2 + D} x \cos x = + (x+3) \sin x - \cos x$$

$$\therefore y = y_c + y_p = A + B e^{-x} + (x+3) \sin x - \cos x$$

iv. solution, given that, $(D^2 - 1)y = x^2 \cos x$

$$\therefore A.E.: m^2 - 1 = 0 \Rightarrow m = \pm 1$$

$$C.F.: y_c = A e^x + B e^{-x}$$

$$P.T.: y_p = \frac{1}{D^2 - 1} x^2 \cos x$$

$$\text{Let } X = \frac{1}{D^2 - 1} x^2 \cos x$$

$$\text{and } Y = \frac{1}{D^2 - 1} x^2 \sin x$$

$$\text{Let now } X + iY = \frac{1}{D^2 - 1} e^{ix} x^2$$

$$= e^{ix} \frac{1}{(D+i)^{r-1}} \cdot x^r$$

$$= e^{ix} \frac{1}{D^r + 2iD - 2} \cdot x^r$$

$$= e^{ix} \cdot \frac{1}{-2 \left\{ 1 - \left(\frac{D^r + 2iD}{2} \right) \right\}} \cdot x^r$$

$$= -\frac{1}{2} e^{ix} \cdot \left\{ 1 - \left(\frac{D^r + 2iD}{2} \right) \right\}^{-1} \cdot x^r$$

$$= -\frac{1}{2} e^{ix} \cdot \left\{ 1 + \left(\frac{D^r + 2iD}{2} \right) + \left(\frac{D^r + 2iD}{2} \right)^2 + \dots \right\} \cdot x^r$$

$$= -\frac{1}{2} e^{ix} \cdot \left\{ 1 + \frac{1}{2} D^r + iD - D^r \right\} x^r$$

$$= -\frac{1}{2} e^{ix} (x^r + 1 + 2ix - 2)$$

$$= -\frac{1}{2} (\cos x + i \sin x) (x^r - 1 + 2ix)$$

$$= -\frac{1}{2} \left[\{ (x^r - 1) \cos x - 2x \sin x \} \right.$$

$$\left. + i \{ (x^r - 1) \sin x + 2x \cos x \} \right]$$

Equating real and imaginary part on both sides

$$x = -\frac{1}{2} \{ (x^r - 1) \cos x - 2x \sin x \}$$

$$= \frac{1}{2} (1 - x^r) \cos x + x \sin x = y_p$$

$$\therefore y = y_c + y_p = A e^x + B e^{-x} + \frac{1}{2} (1 - x^r) \cos x + x \sin x$$

$v, v_i \rightarrow$ For practice

Q. solve the differential equations

i. $(D^2 - 4)y = e^x - \sin x$

ii. $(D^2 - 2D + 3)y = \cos x + x$

iii. $(D^2 + 1)y = e^x x - e^x$

iv. $(D^3 - 3D^2 + 4D - 2)y = e^x + \cos x$

For practice.

Application

Problem-1: A mass m , free to move along the x -axis, is attracted toward the origin with a force proportional to its distance from the origin. Find the motion (a) if it starts from at $x = x_0$ and (b) if it starts at $x = x_0$ with initial velocity v_0 moving away from the origin.

Solution: We know, the differential equation of free undamped motion, $m \frac{d^2x}{dt^2} + kx = 0 \rightarrow (1)$

$$\Rightarrow \frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

$$\Rightarrow \frac{d^2x}{dt^2} + \lambda^2 x = 0 \rightarrow (2) \quad \because \lambda^2 = \frac{k}{m}$$

Let, $x = e^{rt}$ be a trial solution of equation (2)

$$\therefore \text{A.E.: } r^2 + \lambda^2 = 0$$

The general solution of (2) can be written as

$$x = C_1 \sin \lambda t + C_2 \cos \lambda t \quad \rightarrow (3)$$

$$\therefore \frac{dx}{dt} = +\lambda C_1 \cos \lambda t - \lambda C_2 \sin \lambda t \quad \rightarrow (4)$$

(a) When $t=0$, $x=x_0$ and $v=0$. Then $C_1=0, C_2=x_0$

From (3) $\Rightarrow x = x_0 \cos \lambda t$

(b) When $t=0$, $x=x_0$ and $v=v_0$. Then $C_1 = \frac{v_0}{\lambda}$

From (3) $\Rightarrow x = \frac{v_0}{\lambda} \sin \lambda t + x_0 \cos \lambda t \quad \rightarrow (5)$

$$= C \left[\frac{v_0/\lambda}{C} \sin \lambda t + \frac{x_0}{C} \cos \lambda t \right]$$

$$= C [-\sin \phi \cdot \sin \lambda t + \cos \phi \cdot \cos \lambda t]$$

$$= C \cos(\lambda t + \phi)$$

$$= C \cos\left(\sqrt{\frac{k}{m}} t + \phi\right) \rightarrow (6)$$

Here, $-\sin \phi = \frac{v_0/\lambda}{C}$

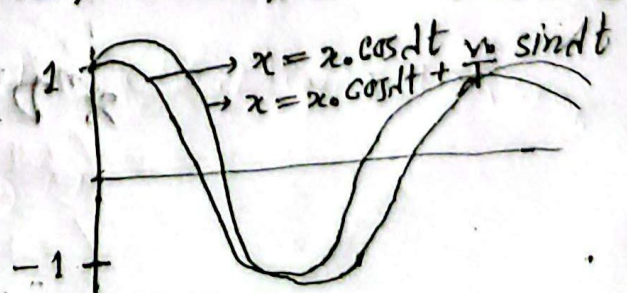
$$\cos \phi = \frac{x_0}{C}$$

$$C^2 = \left(\frac{v_0}{\lambda}\right)^2 + x_0^2$$

From (5) $\Rightarrow$

In (a) the motion is simple harmonic motion of amplitude x_0 and period $2\pi/\lambda$.

In (b) the motion is simple harmonic motion of amplitude $\sqrt{\left(\frac{v_0}{\lambda}\right)^2 + x_0^2}$ and period $2\pi/\lambda$.



Homogeneous Linear Equations with variable Coefficients

*. Write the general form of Cauchy-Euler's linear differential equation.

$$(a_0 x^n D^n + a_1 x^{n-1} D^{n-1} + \dots + a_{n-1} D + a_n) y = 0$$

$$\text{Let, } x = e^z, \quad z = \ln x$$

$$\text{Now, } Dy = \frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \frac{1}{x} \cdot \frac{dy}{dz} \rightarrow (1)$$

$$\Rightarrow x Dy = Dy \quad \left[\because D = \frac{d}{dz} \right]$$

$$\text{Again, } D^2 y = \frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

$$= \frac{d}{dx} \left(\frac{1}{x} \cdot \frac{dy}{dz} \right)$$

$$= \frac{1}{x} \cdot \frac{dy}{dz} \cdot \frac{dz}{dx} - \frac{1}{x^2} \cdot \frac{dy}{dz}$$

$$= \frac{1}{x^2} \left(\frac{dy}{dz} - \frac{dy}{dz} \right) \quad / \because \frac{dz}{dx} = \frac{1}{x}$$

$$= \frac{1}{x^2} (D^2 y - Dy) = \frac{1}{x^2} D(D-1)y$$

$$\Rightarrow x^2 D^2 y = D(D-1)y$$

$$\text{Similarly, } x^3 D^3 y = D(D-1)(D-2)y$$

$$x^n D^n y = D(D-1)(D-2) \dots (D-n+1)y$$

* If $x = e^z$ or $z = \ln x$,
Solve the differential equations

1. $(x^2 D^2 - 6x D + 6) y = 0$

2. $(x^2 D^2 - x D + 1) y = \ln x$

3. $(x^2 D^2 + x D) y = 12 \ln x$

4. $(x^2 D^2 - 2x D + 2) y = 4x$

5. $(x^2 D^2 + x D + 1) y = \sin(\ln x^2)$

Ex. 1. solution: Given that, $(x^2 D^2 - 6x D + 6) y = 0$ $\rightarrow (1)$

Let, $x = e^z \Rightarrow z = \ln x$.

Now, $Dy = \frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \frac{1}{x} \cdot \frac{dy}{dz}$

$$\Rightarrow Dy = \frac{1}{x} Dy \Rightarrow x Dy = Dy$$

Similarly, $x D^2 y = D(D-1)y$

From (1) $\Rightarrow \{D(D-1) - 6D + 6\} y = 0$

$$\Rightarrow (D^2 - 7D + 6) y = 0$$

Let, $y = e^{mz}$ be a trial solution,

$$\therefore \text{A.E.: } m^2 - 7m + 6 = 0$$

$$\therefore m = 1, 6$$

The general solution is, $y = C_1 e^x + C_2 e^{6x}$
 $= C_1 x + C_2 x^6$.

2. solution: Given that, $(x^2 D^2 - xD + 1)y = \ln x \rightarrow (1)$

Let, $x = e^z, \Rightarrow z = \ln x$ and $D = \frac{d}{dz}$

We know, $x D = D$ and $x^2 D^2 = D(D-1)$

$$(1) \Rightarrow \{D(D-1) - D + 1\}y = \ln e^z$$

$$\Rightarrow (D^2 - D - D + 1)y = z$$

$$\Rightarrow (D^2 - 2D + 1)y = z \rightarrow (2)$$

Let, $y = e^{mz}$ be a trial solution of (2).

$$\therefore \text{A.E.: } m^2 - 2m + 1 = 0$$

$$\Rightarrow m = 1, 1$$

$$\therefore \text{C.F.: } y_c = (C_1 + C_2 z) e^z$$

$$= (C_1 + C_2 \ln x) x$$

$$\therefore \text{P.I.: } y_p = \frac{1}{(D^2 - 2D + 1)} \cdot z$$

$$= (1 - D)^{-1} \cdot z$$

$$= (1 + 2D + 3D^2 + \dots) \cdot z$$

$$= (z + 2) = \ln 2 + 2$$

∴ The general solution is, $y = y_c + y_p$

$$= (C_1 + C_2 \ln x) \cdot x + \ln x + 2.$$

3. solution: Given that, $(x^2 D^2 + x D) y = 12 \ln x \rightarrow (1)$

Let, $x = e^z \Rightarrow z = \ln x$, $D = \frac{d}{dz}$

We know, $x D = D$ and $x^2 D^2 = D(D-1)$

$$(1) \Rightarrow \{D(D-1) + D\} y = 12z$$

$$\Rightarrow \{D^2 y = 12z \rightarrow (2)$$

∴ Let, $y = e^{mz}$ be a trial solution of (2).

$$\therefore \text{A.E.: } m^2 = 0 \Rightarrow m = 0, 0$$

$$\therefore y_c = C_1 + C_2 z$$

$$y_p = \frac{1}{D^2} \cdot 12z = 2z^3$$

$$\therefore y = y_c + y_p = C_1 + C_2 z + 2z^3$$

$$= C_1 + C_2 \cdot \ln x + 2(\ln x)^3. \underline{\underline{\text{Ans.}}}$$

4. solution: Given that, $(x^2 D^2 - 2x D + 2)y = 4x \rightarrow (1)$

Let, $x = e^z \Rightarrow z = \ln x$; $D = \frac{d}{dz}$

We know, $x D = D$ and $x^2 D^2 = D(D-1)$

$$(1) \Rightarrow \{D(D-1) - 2D + 2\} y = 4 \ln x e^z$$

$$\Rightarrow (D^2 - 3D + 2)y = 4e^z \rightarrow (2)$$

Let, $y = e^{mz}$ be a trial solution of (2).

$$\therefore \text{A.E.: } m^2 - 3m + 2 = 0$$

$$\Rightarrow m = 1, 2$$

$$\therefore \text{C.F.: } y_c = C_1 e^z + C_2 e^{2z}$$

$$= C_1 x + C_2 x^2$$

$$\text{P.I.: } y_p = \frac{1}{D^2 - 3D + 2} \cdot 4e^z$$

$$= \frac{1}{2D - 3} \cdot 4e^z$$

$$= \frac{1}{2 \cdot 1 - 3} \cdot 4e^z$$

$$= -4z e^z$$

$$= -4x \ln x$$

$\therefore$ The general solution is, $y = y_c + y_p$

$$= C_1 x + C_2 x^2 - 4x \ln x. \quad \underline{\underline{Ans.}}$$

5. Solution: Given that, $(x^2 D^2 + xD + 1)y = \sin(\ln x^2)$

Let, $x = e^z \Rightarrow z = \ln x$, $D = \frac{d}{dz}$ $\longrightarrow (1)$

We know, $x D = D$ and $x^2 D^2 = D(D-1)$

$$(1) \Rightarrow \{ D(D-1) + D + 1 \} y = \sin(\ln(e^z)^2)$$

$$\Rightarrow (D^2 - D + D + 1) y = \sin(\ln e^{2z})$$

$$\Rightarrow (D^2 + 1) y = \sin 2z$$

~~A.E.~~ Let, $y = e^{mz}$ be a trial solution of (2).

$$\therefore \text{A.E.: } m^2 + 1 = 0$$

$$\Rightarrow m = \pm i$$

$$\therefore \text{C.F.: } y_c = C_1 \cos z + C_2 \sin z$$

$$= C_1 \cos(\ln x) + C_2 \sin(\ln x)$$

$$\text{P.I.: } y_p = \frac{1}{D^2 + 1} \sin 2z$$

$$= \frac{1}{-2^2 + 1} \sin 2z$$

$$= -\frac{1}{3} \sin 2z$$

$$= -\frac{1}{3} \sin(2 \ln x)$$

$\therefore$ The general solution is, $y = y_c + y_p$

$$= C_1 \cos(\ln x) + C_2 \sin(\ln x) - \frac{1}{3} \sin(2 \ln x)$$

Application

*** Q. Problem: A spring for which $k = 700 \text{ Nm}^{-1}$ hangs in a vertical position with its upper end fixed. A mass of 7 kg is attached to the lower end. After coming to rest, the mass is pulled down 0.05 m and released. Discuss the resulting motion of the mass, neglecting air resistance.

Solution: Take the origin at the centre of gravity of the mass after coming to rest and let x be the change in position of the mass at time t . When the mass is at rest the spring force is equal to but opposite in direction to the gravitational force. The net force at time t is the spring force $-kx$ corresponding to the change x in the position of the mass. Then

$$m \frac{d^2x}{dt^2} = -kx$$

$$\Rightarrow 7 \frac{d^2x}{dt^2} = -700x$$

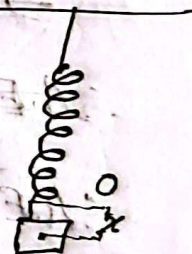
$$\Rightarrow 7 \frac{d^2x}{dt^2} + 700x = 0$$

$$\Rightarrow \frac{d^2x}{dt^2} + 100x = 0 \longrightarrow (1)$$

Here,

$$k = 700 \text{ Nm}^{-1}$$

$$m = 7 \text{ kg}$$



Let, $x = e^{mt}$ be a trial solution of (1).

$$\therefore \text{A.E.} : m^2 + 100 = 0$$

$$\Rightarrow m = \pm 10i$$

$\therefore$ The general solution is, $x = C_1 \sin 10t + C_2 \cos 10t$ $\rightarrow (2)$

$$\text{Now, } \frac{dx}{dt} = 10 (C_1 \cos 10t - C_2 \sin 10t) \rightarrow (3)$$

When $t = 0$, $x = 0.05 \text{ m}$ and $v = 0$, Then

$$C_2 = 0.05, C_1 = 0.$$

$$\text{From (2)} \Rightarrow x = 0.05 \cos 10t.$$

This represents a simple harmonic motion.

The period is $\frac{2\pi}{10} = 0.628 \text{ sec}$ the frequency

is $\frac{10}{2\pi} = 1.59 \text{ cycles/sec}$, and the amplitude is 0.05 m .

Legendre polynomials and functions

* Legendre's differential equation:

$$(1-x^2) \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0 \rightarrow (1)$$

* Legendre polynomial

$$P_n(x) = \sum_{r=0}^n \frac{(-1)^r}{2^n r!} \frac{(2n-2r)!}{(n-r)! (n-2r)!} x^{n-2r}$$

$$* P_0(x) = 1$$

$$P_1(x) = x$$

$$P_2(x) = \frac{1}{2} (3x^2 - 1)$$

$$P_3(x) = \frac{1}{2} (5x^3 - 3x)$$

*** Q. Problem: From the orthogonal theorem of Legendre polynomials, show that

$$\int_{-1}^1 P_m(x) \cdot P_n(x) dx = 0, m \neq n.$$

Solution: We know, the Legendre's differential equation, $(1-x^2) \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0 \rightarrow (1)$

Let, $y = P_m(x)$ & $P_n(x)$ be two solutions of (1),

$$\text{We get } (1-x^2) P_m''(x) - 2x P_m'(x) + m(m+1) P_m(x) = 0 \rightarrow (2)$$

$$\& (1-x^2) P_n''(x) - 2x P_n'(x) + n(n+1) P_n(x) = 0 \rightarrow (3)$$

$$\text{NOW, } (2) \times P_n(x) - (3) \times P_m(x) \Rightarrow$$

$$(1-x^2) [P_m''(x) \cdot P_n(x) - P_n''(x) P_m(x)]$$

$$- 2x [P_m'(x) \cdot P_n(x) - P_m(x) \cdot P_n'(x)]$$

$$+ (m^2 + m - n^2 - n) P_m(x) \cdot P_n(x) = 0$$

$$\Rightarrow \frac{d}{dx} [(1-x^2) \cdot \{ P_m'(x) \cdot P_n(x) - P_m(x) \cdot P_n'(x) \}]$$

$$= - (m-n)(m+n+1) P_m(x) P_n(x)$$

Integrating on both sides with respect to x and limit -1 to 1 , we get

$$\Rightarrow - (m-n)(m+n+1) \int_{-1}^1 P_m(x) P_n(x) dx$$

$$= [(1-x^2) \cdot \{ P_m'(x) \cdot P_n(x) - P_m(x) \cdot P_n'(x) \}]_{-1}^1$$

$$\Rightarrow - (m-n) \cdot (m+n+1) \int_{-1}^1 P_m(x) \cdot P_n(x) dx = 0$$

$$\Rightarrow \int_{-1}^1 P_m(x) \cdot P_n(x) dx = 0. \quad \underline{\text{(shown)}}$$

math

A semi-infinite solid $x > 0$ is initially at temperature zero. At time $t = 0$, a constant temperature u_0 is applied and maintained at the face $x = 0$. Find the temperature at any point of the solid and at any time $t > 0$.

Solution we know, the one-dimensional heat equation, $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$, $x > 0, t > 0$

The given condition, $u(x, 0) = 0$ ($x > 0$).

$$u(0, t) = u_0 \quad (t > 0)$$

Let, similarity variable $\eta = \frac{x}{2\sqrt{kt}}$ and

$$u(x, t) = u_0 f(\eta)$$

Now, the ordinary differential equation

$$f'' + 2\eta f' = 0, \quad f(0) = 1, \quad f(\infty) = 0$$

$\therefore$ the complementary error function

$$u(x, t) = u_0 \operatorname{erf}\left(\frac{x}{2\sqrt{kt}}\right)$$

$\therefore$ We can say, As time increases, the heat diffuses inward from the surface at $x = 0$. The erf factor shows that near the surface

$u \approx u_0$ while far away $u \rightarrow 0$, and the scale of diffusion is $\sqrt{xt}$.

* Legendre polynomials,

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2 - 1)$$

$$P_3(x) = \frac{1}{2}(5x^3 - 3x)$$

* Express $f(x) = 4x^3 + 6x^2 + 7x + 2$ in terms of Legendre polynomials.

Solution: Let,

$$4x^3 + 6x^2 + 7x + 2 = aP_3(x) + bP_2(x) + cP_1(x) + dP_0(x) \longrightarrow (1)$$

$$= a \left(\frac{5}{2}x^3 - \frac{3}{2}x \right) + b \left(\frac{3}{2}x^2 - \frac{1}{2} \right) + cx + d$$

$$= \frac{5a}{2}x^3 + \frac{3b}{2}x^2 + \left(-\frac{3a}{2} + c \right)x - \frac{b}{2} + d$$

Equating on both sides,

$$4 = \frac{5}{2}a \Rightarrow a = \frac{8}{5}$$

$$6 = \frac{3b}{2} \Rightarrow b = 4$$

$$7 = \frac{-3a}{2} + c \Rightarrow c = 7 + \frac{3}{2} \cdot \frac{8}{5} = \frac{47}{5}$$

$$2 = \frac{-b}{2} + d \Rightarrow 2 + \frac{4}{2} = d \Rightarrow d = 4$$

putting the values of a, b, c and d in (1), we get

$$4x^3 + 6x^2 + 7x + 2 = \frac{8}{5}P_3(x) + 4P_2(x) + \frac{47}{5}P_1(x)$$

$$+ 4P_0(x) \quad \underline{\underline{\text{Ans}}}$$

$$2x^3 + 5x^2 + 7x + 2 = (x)^3 + 2x^2 + 3x + 2$$

$$(x)^3 + 2x^2 + 3x + 2 = (x)^3 + 2x^2 + 3x + 2$$

$$(x)^3 + 2x^2 + 3x + 2 = (x)^3 + 2x^2 + 3x + 2$$

$$(x)^3 + 2x^2 + 3x + 2 = (x)^3 + 2x^2 + 3x + 2$$

$$(x)^3 + 2x^2 + 3x + 2 = (x)^3 + 2x^2 + 3x + 2$$

$$b + x^2 +$$

$$x^3 + \frac{2x^2}{2} + x \frac{2x}{2} +$$

$$b + \frac{2}{2} =$$

Math
***, other method:

We know, the one dimensional heat equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad x > 0, t > 0. \rightarrow (1)$$

$$u(x, 0) = 0, \quad u(0, t) = u_0, \quad u(x, t) \rightarrow 0 \quad (x \rightarrow \infty)$$

Take the Laplace transform in t :

$$L\{u(x, t)\} = \int_0^\infty e^{-st} u(x, t) dt$$

Since, $u(x, 0) = 0$, the transform of the Partial Differential equations gives,

$$su = c^2 u_{xx}$$

$$\Rightarrow u_{xx} - \frac{s}{c^2} u = 0$$

$$\therefore u(x, s) = A e^{-\frac{\sqrt{s}}{c} x} + B e^{\frac{\sqrt{s}}{c} x}$$

When $s \rightarrow \infty$, then $B = 0$ and $u(0, t)$ then $A = \frac{u_0}{s}$.

$$\therefore u(x, s) = \frac{u_0}{s} e^{-\frac{\sqrt{s}}{c} x}$$

Applying Laplace inverse transformation,

$$\therefore u(x, t) = u_0 \operatorname{erf}\left(\frac{x}{2c\sqrt{t}}\right)$$